

Lab 1: Vulnerability Analysis

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Upon the start of this lab, we were tasked with exploring the various vulnerability analysis tools provided on the Virtual Machines, specifically host and network vulnerability analysis. There are several tools available on both Windows and Linux that are used for vulnerability analysis on networks and can be very successful. Knowing what tool to use is extremely important for analysis. "Vulnerability assessment is an evaluation method that enables organizations to review their systems for potential security weaknesses" (Fortinet, n.d.).

For Windows, I chose Nmap to do the analysis. "Nmap ("Network Mapper") is a free and open source utility for network discovery and security auditing...Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics" (nmap, 2017). Nmap is open source and free which increases its accessibility as a vulnerability tool. I have used nmap in the past, but it had been a little bit so I had to do research on the best way to run a scan. Luckily, nmap has a helpful section built in that lists all of the different scans that can be done, Figure 1. Using this, I picked out a few that might be useful when looking at security.

I ran a scan on mandia_ftp_3 from Windows, starting off with the command "nmap -sn 172.16.3.242" to make sure that it was up. -sn is a ping scan. This let me know that the host was up so I would be able to continue using nmap on it, Figure 2. Next, I used the command "nmap -sS 172.16.3.242" to do a TCP SYN scan, Figure 3. This shows the TCP ports that are open on the target machine. The reason this is an important scan to use, is because while these ports are used for network communication, they can also be used for unauthorized access to the machine. Being aware of what ports are open and what ports are known for being exploited. On top of that, any ports that are unnecessary to have open can be closed to improve security. Next, I used "nmap -sA 172.16.3.242" which is a TCP ACK port scan, Figure 4. This revealed that all of the ports on this machine are unfiltered. "Nmap cannot determine whether the port is open because packet filtering prevents its probes from reaching the port. The filtering could be from a dedicated firewall device, router rules, or host-based firewall software" (Lyon, 2020).

For the Kali Linux VM, I chose to use Sparta for scanning. "SPARTA is a python GUI application which simplifies network infrastructure penetration testing by aiding the penetration tester in the scanning and enumeration phase" (Quina & Stavliotis, 2023). Sparta combines several tools used for analysis and centralizes them into a singular application. This reduces time and effort needed for testing and in my opinion, likely increases accuracy to degree. SPARTA is also only on Linux which removes usability from Windows users. "Sparta integrates a variety of vulnerability scanning tools, such as Nessus, OpenVAS, and Nikto, directly into its interface" (Cyberly, 2025). SPARTA is

also open source and free which provides an opportunity for anyone to use the tool, so even smaller companies are able to use the tool.

When running the scan using SPARTA, I added `mandia_ftp_3` as the host, using its IP address, Figure 5. Upon adding Mandia, it automatically started to scan, because of the settings shown in Figure 5 that say “Run nmap host discovery” and “Run staged nmap scan”. The scan takes a little bit of time, but not too long. As SPARTA is actively scanning, it shows a progress bar, Figure 6. I find this preferable to nmap on its own since sometimes it’s hard to tell if nmap is actually doing something considering it’s a command line application. Figure 7 shows the output once the scans are complete. This reveals what ports are open, what the name of it is, and the version. SPARTA also shows information on the host, Figure 8. It gives a summary of the device including IPv4, MAC, open ports, closed ports, and filtered ports. This summary provides a good overview and a base understanding of security on the machine.

Next, I returned to the Windows VM to run nmap again but against the Windows XP VM, `win-xp-3`. This VM is not shown as deployed in our virtual machine network, so I was unsure on if a scan would actually work, Figure 9. I decided to run the same scans I had previously for the Mandia VM for consistency purposes. So the first command I entered was “`nmap -sn 172.16.3.213`” to make sure that the VM was up. The result was less than shocking, as nmap said that the host was down, Figure 10. Despite this, I did attempt to run the other commands used previously, however none of them went through since the host was still down, Figure 11. With this, there was no other information that I could gather from it at that time.

I moved back to the Kali VM to scan `win-hunt-3` that had been used for the scans previously. I reopened SPARTA and made the Windows machine the host now instead of the Mandia machine, Figure 12. After letting it run for a while, SPARTA was finished with the scan, Figure 13. This revealed that there were 13 ports open on the host, 0 closed ports, and 65,522 filtered ports.

Between all of the scans, there were quite a few glaring differences, in my opinion. For these comparisons, I will be excluding the Windows XP machine since it was not running and did not return any other information besides that. Starting off, the `mandia_ftp_3` came back as Microsoft Windows Longhorn. “The development of Windows Vista (codenamed Longhorn) began in May 2001, prior to the completion of Microsoft's Windows XP operating system, and continued until November 8, 2006, when it was released to manufacturing” (Wikipedia, 2024). This explains why there are more ports open by default on this machine. Security was not as developed when Vista was released and likely had less strict firewall rules, more services that utilized various ports, and protocols that were discontinued at a later date. Having these ports open wasn’t considered a security risk because of the lack of foresight at this time. `Mandia_ftp_3` also had way more closed ports than `win-hunt-3`. This is likely because Windows 2008 had stricter firewall filtered port rules. The majority of the ports that are open on both of

these machines are TCP ports. “Devices use TCP ports to identify specific applications and services, ensuring network traffic reaches the correct destination. Generally, a TCP port represents an application—or service-specific endpoint identifier” (Heintzkill, 2021).

I chose both SPARTA and nmap because they are tools that I have used in the past. I wanted to brush up on them because it had been a little bit since I had used them. Both of these tools are extremely useful in security assessments as they can tell a lot about potential network traffic and devices. This can help with preventative security and prepare systems before a threat occurs. Between the two, I would say SPARTA is the better option because nmap is integrated into it, along with a couple other tools. SPARTA also has a GUI that provides a cleaner look to the output and makes it more organized.

The weaknesses of SPARTA lie in the fact that it is Linux exclusive. This means that organizations using Windows aren’t able to use it on their systems. Windows systems still have access to nmap so there are still options available. After reviewing SPARTA’s GitHub, it is also clear that it isn’t really in active development and only gets updates about once a year. “Sparta automates network discovery by running scans to identify live hosts, open ports, and services running on a target network. By performing comprehensive port scanning, Sparta reveals the entire attack surface of the target system” (Cyberly, 2025). This provides a serious advantage when it comes to vulnerability analysis and provides a much larger vision of what could be going on or potential threats.

The downside of nmap is that it’s a command line application so users have to know what to input to get results. Luckily, nmap has a very helpful section that goes over all of the commands that can be used and what they do. However, SPARTA automatically chooses the best nmap options and makes it simpler. Nmap also takes a lot of time to run and in the command line, it can seem like nothing is happening. With SPARTA the visual progress bar allows users to know that the application is still running, it’s just taking time. “One of the biggest limitations of Nmap is its vulnerability to detection by security systems like firewalls, intrusion detection systems (IDS), and intrusion prevention systems (IPS). These systems are specifically designed to identify and block malicious or unauthorised network traffic” (Cyberly, 2024). This makes the use of nmap obsolete in some cases and should be taken into account when doing vulnerability analysis.

Overall, this lab provided a good insight to a couple tools that can help with vulnerability analysis. Getting to explore different versions of Windows also showed the evolution of Windows systems as security improved. This information will be valuable to keep in mind when having to set up devices on a network or when having to prepare security for a device. Knowing vulnerabilities of a system is the first step in vulnerability analysis.

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Appendices

Figure 1: Nmap help section

```

Administrator: C:\Windows\system32\cmd.exe

C:\Program Files (x86)\Nmap>nmap.exe
Nmap 7.70 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] [target specification]

TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iI <inputfilename>: Input from list of hosts/networks
  -IR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file

HOST DISCOVERY:
  -sn: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PV/PV[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host

SCAN TECHNIQUES:
  -sS/sT/sA/sM/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sM/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sV/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan

PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -pU:55,111,137,T:21-25,80,139,8080,519
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -i: Scan ports consecutively - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>

SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)

SCRIPT SCAN:
  -sC: equivalent to --script=default
  --script <lua scripts>: <lua scripts> is a comma separated list of
    directories, script-files or script-categories
  --script-args <n1=v1[,n2=v2,...]>: provide arguments to scripts
  --script-args-file <filename>: provide NSE script args in a file
  --script-trace: Show all data sent and received
  --script-updatedb: Update the script database.
  --script-help <lua scripts>: Show help about scripts.
    <lua scripts> is a comma-separated list of script-files or
    script-categories.

OS DETECTION:
  -O: Enable OS detection
  --oscan-limit: Limit OS detection to promising targets
  --osscan-guess: Guess OS more aggressively

TIMING AND PERFORMANCE:
  Options which take <time> are in seconds, or append 'ms' (milliseconds),
  's' (seconds), 'm' (minutes), or 'h' (hours) to the value (e.g. 30m).
  -T<0-5>: Set timing template (higher is faster)
  --min-hostgroup/max-hostgroup <size>: Parallel host scan group sizes
  --min-parallelism/max-parallelism <numprobes>: Probe parallelization
  --min-rtt-timeout/max-rtt-timeout/initial-rtt-timeout <time>: Specifies
    probe round trip time.
  --max-retries <tries>: Caps number of port scan probe retransmissions.
  --host-timeout <time>: Give up on target after this long
  --scan-delay/--max-scan-delay <time>: Adjust delay between probes
  --min-rate <number>: Send packets no slower than <number> per second
  --max-rate <number>: Send packets no faster than <number> per second

FIREWALL/IDS EVASION AND SPOOFING:
  -f; --mtu <val>: Fragment packets (optionally w/given MTU)
  -D <decv1,decv2[:DF]...>: Cloak a scan with decoys

```

Figure 2: Nmap reporting that mandia_ftp_3 is up

```

C:\Program Files (x86)\Nmap>nmap -sn 172.16.3.242
Starting Nmap 7.70 ( https://nmap.org ) at 2025-09-17 14:31 Eastern Daylight Time
Nmap scan report for mandia_ftp_3.ISCS-int.lan (172.16.3.242)
Host is up (0.00s latency).
MAC Address: 00:02:B3:00:0E:05 (Intel)
Nmap done: 1 IP address (1 host up) scanned in 0.78 seconds

C:\Program Files (x86)\Nmap>_

```

Figure 3: Nmap showing open ports

```
C:\Program Files (x86)\Nmap>nmap -sS 172.16.3.242
Starting Nmap 7.70 ( https://nmap.org ) at 2025-09-17 14:47 Eastern Daylight Time
Nmap scan report for mandia_ftp_3.ISCS-int.lan (172.16.3.242)
Host is up (0.00s latency).
Not shown: 988 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
1801/tcp   open  msmq
2103/tcp   open  zephyr-clt
2105/tcp   open  eklogin
2107/tcp   open  msmq-mgmt
3389/tcp   open  ms-wbt-server
8080/tcp   open  http-proxy
8888/tcp   open  sun-answerbook
MAC Address: 00:02:B3:00:0E:05 (Intel)

Nmap done: 1 IP address (1 host up) scanned in 2.06 seconds
```

Figure 4: TCP ACK port scan

```
C:\Program Files (x86)\Nmap>nmap -sA 172.16.3.242
Starting Nmap 7.70 ( https://nmap.org ) at 2025-09-17 19:40 Eastern Daylight Time
Nmap scan report for mandia_ftp_3.ISCS-int.lan (172.16.3.242)
Host is up (0.00s latency).
All 1000 scanned ports on mandia_ftp_3.ISCS-int.lan (172.16.3.242) are unfiltered
MAC Address: 00:02:B3:00:0E:05 (Intel)

Nmap done: 1 IP address (1 host up) scanned in 2.16 seconds

C:\Program Files (x86)\Nmap>_
```

Figure 5: Adding Mandia as the host to scope in SPARTA

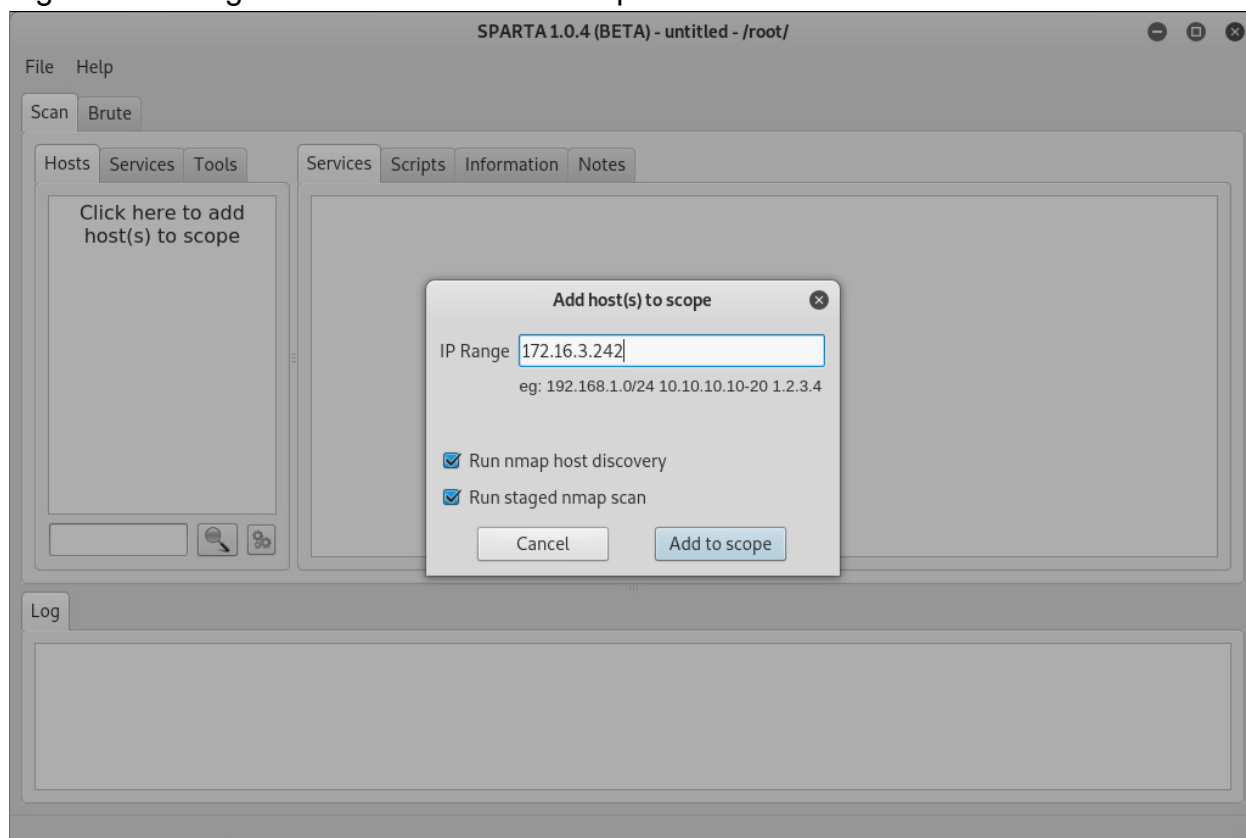


Figure 6: SPARTA progress bars

Progress	Tool	Host	Start time	End time	Status
	smbenum (445/tcp)	172.16.3.242	18 Sep 2025 10:49:36	18 Sep 2025 10:49:39	Finished
	nmap (stage 3)	172.16.3.242	18 Sep 2025 10:49:36		Running
	screenshot (80/tcp)	172.16.3.242	18 Sep 2025 10:49:18	18 Sep 2025 10:49:18	Finished
	nikto (80/tcp)	172.16.3.242	18 Sep 2025 10:49:17	18 Sep 2025 10:49:29	Finished
	nmap (stage 2)	172.16.3.242	18 Sep 2025 10:49:17	18 Sep 2025 10:49:36	Finished
	nmap (stage 1)	172.16.3.242	18 Sep 2025 10:49:11	18 Sep 2025 10:49:17	Finished

Figure 7: SPARTA Scan results

Port	Protocol	State	Name	Version
21	tcp	open	ftp	Microsoft ftpd
80	tcp	open	http	Microsoft IIS httpd 10.0
135	tcp	open	msrpc	Microsoft Windows RPC
137	udp	open	netbios-ns	Microsoft Windows Mobile netbios-ns
139	tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445	tcp	open	microsoft-ds	
1801	tcp	open	msmq	
2103	tcp	open	msrpc	Microsoft Windows RPC
2105	tcp	open	msrpc	Microsoft Windows RPC
2107	tcp	open	msrpc	Microsoft Windows RPC
3389	tcp	open	ms-wbt-ser...	Microsoft Terminal Services
5040	tcp	open	unknown	
8080	tcp	open	http-proxy	
8888	tcp	open	sun-answer...	
49664	tcp	open	msrpc	Microsoft Windows RPC
49665	tcp	open	msrpc	Microsoft Windows RPC
49666	tcp	open	msrpc	Microsoft Windows RPC
49667	tcp	open	msrpc	Microsoft Windows RPC
49668	tcp	open	msrpc	Microsoft Windows RPC
49669	tcp	open	msrpc	Microsoft Windows RPC
49670	tcp	open	msrpc	Microsoft Windows RPC
49671	tcp	open	msrpc	Microsoft Windows RPC
49672	tcp	open	msrpc	Microsoft Windows RPC

Figure 8: Information from SPARTA

Services

Scripts

Information

Notes

nikto (80/tcp) 

screenshot (80/t

Host Status

State: up

Open Ports: 23

Closed Ports: 65511

Filtered Ports: 1

Addresses

IPv4: 172.16.3.242

IPv6:

MAC: 00:02:B3:00:0E:05

Operating System

Name: Microsoft Windows Longhorn

Accuracy: 95

Figure 9: Windows XP machines not deployed

☐ Select All Virtual Machines				
☐ win-xp-1	 0	Not Deployed	No Description	
☐ win-xp-10	 0	Not Deployed	No Description	
☐ win-xp-11	 0	Not Deployed	No Description	
☐ win-xp-12	 0	Not Deployed	No Description	
☐ win-xp-13	 0	Not Deployed	No Description	
☐ win-xp-14	 0	Not Deployed	No Description	
☐ win-xp-15	 0	Not Deployed	No Description	
☐ win-xp-16	 0	Not Deployed	No Description	
☐ win-xp-17	 0	Not Deployed	No Description	
☐ win-xp-18	 0	Not Deployed	No Description	
☐ win-xp-19	 0	Not Deployed	No Description	
☐ win-xp-2	 0	Not Deployed	No Description	
☐ win-xp-20	 0	Not Deployed	No Description	
☐ win-xp-21	 0	Not Deployed	No Description	
☐ win-xp-22	 0	Not Deployed	No Description	
☐ win-xp-23	 0	Not Deployed	No Description	
☐ win-xp-24	 0	Not Deployed	No Description	
☐ win-xp-25	 0	Not Deployed	No Description	
☐ win-xp-3	 0	Not Deployed	No Description	
☐ win-xp-4	 0	Not Deployed	No Description	
☐ win-xp-5	 0	Not Deployed	No Description	
☐ win-xp-6	 0	Not Deployed	No Description	
☐ win-xp-7	 0	Not Deployed	No Description	
☐ win-xp-8	 0	Not Deployed	No Description	
☐ win-xp-9	 0	Not Deployed	No Description	

Figure 10: Nmap scan not working because host is down

```

C:\Program Files (x86)\Nmap>nmap -sn 172.16.3.213
Starting Nmap 7.70 ( https://nmap.org ) at 2025-09-18 11:46 Eastern Daylight Time
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.08 seconds

C:\Program Files (x86)\Nmap>

```

Figure 11: Commands not going through for nmap

```
C:\Program Files (x86)\Nmap>nmap -Pn 172.16.3.213
Starting Nmap 7.70 ( https://nmap.org ) at 2025-09-18 11:50 Eastern Daylight Time
Nmap done: 1 IP address (0 hosts up) scanned in 1.06 seconds

C:\Program Files (x86)\Nmap>nmap -sS 172.16.3.213
Starting Nmap 7.70 ( https://nmap.org ) at 2025-09-18 12:07 Eastern Daylight Time
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.17 seconds

C:\Program Files (x86)\Nmap>nmap -sA 172.16.3.213
Starting Nmap 7.70 ( https://nmap.org ) at 2025-09-18 12:08 Eastern Daylight Time
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.08 seconds
```

Figure 12: SPARTA set up for Win-hunt-3

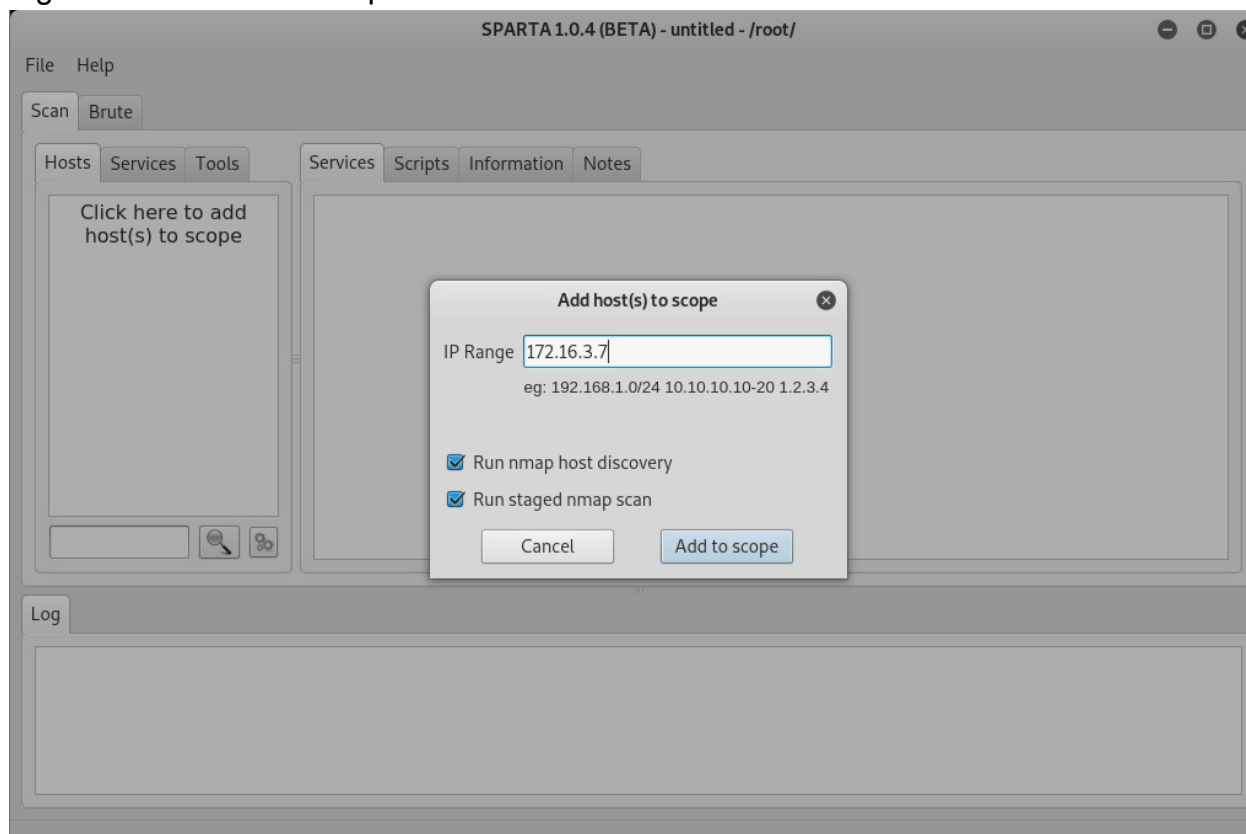


Figure 13: Output from SPARTA based on win-hunt-3

The screenshot shows the SPARTA application interface. The top menu bar includes 'Scan' and 'Brute'. Below it, a tabbed interface shows 'Hosts', 'Services', and 'Tools'. The 'Hosts' tab is active, displaying a list of hosts. The first host, '172.16.3.7 (win-hunt-3)', is selected. The 'Services' tab is also visible, showing a list of services. The 'Scripts' tab is active, displaying a list of scripts. The 'Information' tab is also visible, showing a list of information. The 'Notes' tab is also visible, showing a list of notes. The 'smbenum (445/tcp)' script is selected. The main display area shows the output of the scan, which is a table of open ports and services.

OS	Host	Port	Protocol	State	Name	Version
	172.16.3.7 (win-hunt-3)	135	tcp	open	msrpc	Microsoft Windows RPC
		137	udp	open	netbios-ns	Microsoft Windows netbios-ns (workgroup: ISCS-INT)
		139	tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
		445	tcp	open	microsoft-ds	
		5040	tcp	open	unknown	
		49664	tcp	open	msrpc	Microsoft Windows RPC
		49665	tcp	open	msrpc	Microsoft Windows RPC
		49666	tcp	open	msrpc	Microsoft Windows RPC
		49667	tcp	open	msrpc	Microsoft Windows RPC
		49668	tcp	open	msrpc	Microsoft Windows RPC
		49669	tcp	open	msrpc	Microsoft Windows RPC
		49670	tcp	open	msrpc	Microsoft Windows RPC
		49671	tcp	open	msrpc	Microsoft Windows RPC